Solution: Problem Challenge 1: Find the Corrupt Pair

Problem Statement

We are given an unsorted array containing ‘n’ numbers taken from the range 1 to ‘n’. The array originally contained all the numbers from 1 to ‘n’, but due to a data error, one of the numbers got duplicated which also resulted in one number going missing. Find both these numbers.

Example 1:

```
Input: [3, 1, 2, 5, 2]
Output: [2, 4]
Explanation: '2' is duplicated and '4' is missing.
```

Example 2:

```
Input: [3, 1, 2, 3, 6, 4]
Output: [3, 5]
Explanation: '3' is duplicated and '5' is missing.
```

Solution

This problem follows the **Cyclic Sort** pattern and shares similarities with Find all Duplicate Numbers. Following a similar approach, we will place each number at its correct index. Once we are done with the cyclic sort, we will iterate through the array to find the number that is not at the correct index. Since only one number got corrupted, the number at the wrong index is the duplicated number and the index itself represents the missing number.

Code

Here is what our algorithm will look like:

Java

```java
class Solution {

  // Function to find two numbers that are swapped in the given array
  public static int[] findNumbers(int[] nums) {
    int i = 0;
    while (i < nums.length) {
      // If the current number is not in its correct position, swap it
      if (nums[i] != nums[nums[i] - 1])
        swap(nums, i, nums[i] - 1);
      else
        i++;
    }

    // Iterate through the array to find the first misplaced number
    for (i = 0; i < nums.length; i++)
      if (nums[i] != i + 1)
        return new int[] { nums[i], i + 1 };

    // If no numbers are swapped, return [-1, -1]
    return new int[] { -1, -1 };
  }
```

Run Code Result | Stdout

You must run your code first

Execute

Time Complexity

The time complexity of the above algorithm is O(n).

Space Complexity

The algorithm runs in constant space O(1).

← Previous

Problem Challenge 1: Find The Corrupt Pair (Easy)

Next →

Problem Challenge 2: Find The Smallest Missing Positive Number (Medium)

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